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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : CS-602

COMPUTER NETWORKS

Time Allotted: 3 Hours

Full Marks: 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

Group – A

(Multiple Choice Type Questions)

1. Choose the correct alternative for *any ten* of the following: 1×10=10

(i) The packet of information at the application layer is called

(a) packet (b) message

(c) segment (d) frame

(ii) HDLC (High-Level Data Link Control) is a

(a) bit oriented protocol (b) byte oriented protocol

(c) both (a) and (b) (d) None of these

(iii) If subnet mask is 255.255.252.0, then how many subnets are available?

(a) 2 (b) 18

(c) 1 (d) 24

(iv) Connection establishment in TCP involves a _____ handshake.

(a) one-way (b) two-way

(c) three-way (d) None of these

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(v) Port number in packet indicates

(a) LAN card port number in a computer.

(b) Host identification number in network.

(c) Unique number for a communication process.

(d) PID number of a communicating process under OS.

(vi) Router solicitation and advertisement message is used by

(a) IP

(b) ARP

(c) ICMP

(d) DHCP

(vii) At which layer circuit switching takes place?

(a) Transport layer

(b) Data link layer

(c) Physical layer

(d) None of the above

(viii) The total number of links required to connect 'n' devices using Mesh Topology is

(a) 2^n

(b) $n(n+1)/2$

(c) $n(n-1)/2$

(d) n^2

(ix) Which of the following is an application layer service?

(a) FTP

(b) Remote login

(c) Mail service

(d) All of these

(x) When host knows its IP address but not its physical address, it can use

(a) RARP

(b) ARP

(c) ICMP

(d) IGMP

(xi) Segmentation is done in

(a) physical layer

(b) data link layer

(c) network layer

(d) transport layer

(xii) Which class of IP address is reserved for multicast communication?

(a) Class A

(b) Class B

(c) Class C

(d) Class D

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Group – B

(Short Answer Type Questions)

Answer *any three* of the following: 5×3=15

2. Compare and contrast between OSI and TCP layered models.

3. Differentiate between CSMA/CD and CSMA/CA.

4. Generate the CRC code for the data word of 1010011110. The divisor is 1011.

5. (a) What is the purpose of the TTL (Time to live) field in the IP header?

   (b) If the IP header is 28 bytes long, what will be the value of the 'HLEN' field (in binary)?

   (c) Write the advantage of ICMP over the IPv 4. 2+2+1=5

6. (a) Why is medium access control technique required?

   (b) What is bit stuffing and character stuffing? 2+3=5

Group – C

(Long Answer Type Questions)

Answer *any three* of the following: 15×3=45

7. (a) Derive the expression of the efficiency of pure ALOHA.

   (b) Compare performance of pure ALOHA with slotted ALOHA.

   (c) What is the minimum window size required for selective repeat ARQ protocol and how?

   (d) What is polling?

   (e) How does CSMA/CD differ from CSMA/CA? 3+3+4+2+3=15

8. (a) How does Go-Back-N ARQ differ from Selective Repeat ARQ?

   (b) A computer is using the following sequence numbers. What is the size of the window?

   0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3, 4, ...

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(c) What does the number on a NAK frame mean for Selective Repeat ARQ? Give one example.

(d) Apply CRC algorithm, to determine the checksum and the transmitted frame for the frame 11010111 and for the generator polynomial $x^3 + x^2 + 1$. 3+3+4+5=15

9. A Class-B network has on the Internet has a subnet mask of 255.255.240.0. What is the number of hosts per subnet? Suppose that you have been assigned an IP address 132.128.0.0 and you wish to have 9 subnets in your organization. What could be the subnet addresses? In CIDR, the prefix can be of any length unlike fixed 8, 16 or 24 in Classful addressing for Class A, B or C respectively. What is its use? 4+6+5=15

10. (a) What is congestion?

(b) Why does congestion occur?

(c) Explain Leaky Bucket algorithm for congestion control.

(d) State the basic difference between TCP and UDP.

(e) Compare IPv4 and IPv6. 2+2+4+3+4=15

11. Write short notes on *any three* of the following: 5×3=15

(a) IP Datagram

(b) Virtual Circuit Switching

(c) LAN Topologies

(d) IP Addressing

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